4.4 Wallis, Beta and the Factorial of a Half

WORKSHEET

The question

STEP II 1995, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What are u_0 and u_1? And roughly what does the graph of $\sin^n t$ look like on $[0, \frac{\pi}{2}]$ when n is large?

STAGE 1 A reduction formula

Ask yourself Integrate $\int \sin t \cdot \sin^{n-1} t \, dt$ by parts, integrating $\sin t$ and differentiating $\sin^{n-1} t$. The new integrand contains $\cos^2 t$; how can you get back to powers of $\sin t$?

STAGE 2 An invariant

Ask yourself The relation says the quantity $nu_n u_{n-1}$ does not change when n goes down by one. What is it equal to at $n = 1$?

STAGE 3 **A squeeze**

Pause and try Use $u_n < u_{n-1}$ and $nu_nu_{n-1} = \frac{1}{2}\pi$ to trap nu_n^2.

What's really going on

Wallis's product

Running the reduction formula down to u_0 or u_1 gives explicit values:

$$u_{2m} = \frac{1\cdot 3\cdot 5\cdots(2m-1)}{2\cdot 4\cdot 6\cdots 2m}\cdot\frac{\pi}{2}, \qquad u_{2m+1} = \frac{2\cdot 4\cdot 6\cdots 2m}{3\cdot 5\cdot 7\cdots(2m+1)}.$$

The squeeze shows $u_{2m}/u_{2m+1} \to 1$. Writing out that ratio and rearranging gives Wallis's product,

$$\frac{\pi}{2} = \lim_{m\to\infty} \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdots\frac{2m\cdot 2m}{(2m-1)(2m+1)}.$$

John Wallis found it in 1655 by a daring process of interpolation, guessing what $\int_0^1 (1-x^2)^{1/2}\,dx$ "should" be by fitting it into a table of integrals with whole-number exponents. It was one of the first infinite products in mathematics, after Viète's (Section 2.5), and the first to use only rational numbers.

The factorial of a half

Wallis's interpolation idea was made precise by Euler in 1729. He found a function, now called the *Gamma function*,

$$\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt,$$

which satisfies $\Gamma(s+1) = s\,\Gamma(s)$ (integrate by parts) and $\Gamma(n+1) = n!$ for whole numbers n. It extends the factorial smoothly to all positive real numbers. The closely related *Beta function*

$$\mathrm{B}(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

includes the integrals of Section 4.3 and Section 2.9, and, after the substitution $x = \sin^2 t$, our u_n. In particular, $\int_0^1 x^{1/2}(1-x)^{1/2}\,\mathrm{d}x = \frac{\pi}{8}$ (exercise 2), which gives $\Gamma(\frac{3}{2})^2/\Gamma(3) = \frac{\pi}{8}$, that is, $\left(\frac{1}{2}\right)! = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$. Equivalently, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which is the same fact as the famous Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$, the reason π appears in the formula for the normal distribution.

Stirling's formula

Stirling's formula says $n! \approx \sqrt{2\pi n}\,(n/e)^n$. The $(n/e)^n$ part is relatively easy to obtain, but the constant $\sqrt{2\pi}$ is harder, and the standard way to find it is exactly this question: the explicit values of u_{2m} involve $(2m)!/(m!)^2$, and the limit $nu_n^2 \to \frac{\pi}{2}$ pins down the constant. A completely different route to the same constant, through probability, is in Section 12.23.

Try this next

1 **Wallis's product.** Using the explicit values of u_{2m} and u_{2m+1}, and $u_{2m+1} < u_{2m} < u_{2m-1}$, prove Wallis's product.

2 **A Beta integral.** Try STEP II 2013, Question 2 (step.maths.org/questions/13-s2-q2). It evaluates $\int_0^1 x^n(1-x)^n\,\mathrm{d}x$ with a reduction formula, then asks what the answer suggests when $n = \frac{1}{2}$: a first glimpse of the factorial of a half.

3 **The Beta function.** Try STEP III 2025, Question 1 (in OCR's download of the 2025 papers, www.ocr.org.uk/Images/750486-2025-past-papers.zip). It develops the general Beta integral $\int_0^1 x^{p-1}(1-x)^{q-1}\,\mathrm{d}x$, its symmetry and a recurrence, and uses them to evaluate an integral over $[0,\infty)$.

4 **Central binomial coefficients.** HARDERShow that $u_{2m} = \binom{2m}{m}\dfrac{\pi}{2^{2m+1}}$, and use $nu_n^2 \to \frac{\pi}{2}$ to deduce that $\binom{2m}{m} \approx \dfrac{4^m}{\sqrt{\pi m}}$ for large m. Check this for $m = 10$.

Further reading

- **Julian Havil, *Gamma*** (Princeton University Press). Includes the Gamma function and its history.
- **Philip J. Davis, "Leonhard Euler's integral: a historical profile of the Gamma function"**, *American Mathematical Monthly* 66 (1959). A classic, prize-winning article, readable with this section behind you.
- **Emil Artin, *The Gamma Function*** (Dover). A short, beautiful monograph for when you have done some analysis.

The full guided solution is in Section 4.4:
https://beyondstep.keithmcnulty.org/ch04/wallis-beta.html